

Chemical Changes

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Evidence of a chemical change can include color change, gas formation, precipitate formation, and temperature or light change when those changes result from a reaction.
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Round 1 - Retrieve It

1. Name four common signs that a chemical reaction may have occurred.

2. What is a precipitate?

3. Does every color change prove a chemical reaction?

4. Why is boiling water not automatically evidence of a chemical change?

Round 2 - Sketch It

Create a four-box "reaction evidence" chart for color, gas, precipitate, and temperature/light. Add a small example or symbol to each.

Chemical Changes - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A metal rusts and forms a reddish-brown coating. Physical or chemical change?

2. Ice melts into liquid water. Physical or chemical change?

3. Two clear solutions are mixed and a solid appears. What evidence is present?

Round 4 - Reason from Evidence

Why is it stronger to use more than one piece of evidence before claiming a chemical reaction occurred?

Chemical Changes - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Chemical Changes in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect chemical changes to conservation of mass. If new substances form, what should happen to total mass in a closed system?

E**Extend**

A sealed bag becomes cold and inflates after two substances are mixed. What evidence suggests a reaction, and what additional observation would you seek?

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Chemical Changes

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Color change, gas forms, precipitate forms, temperature or light change.
2. A new solid that forms from substances in solution.
3. No. It is evidence that must be interpreted with the situation.
4. The gas is still water; no new substance is formed.

Sketch It - look for

Any accurate, labeled examples showing the four evidence categories. Examples should distinguish reaction evidence from simple phase change where possible.

Apply It

1. Chemical change; a new substance forms and color changes.
2. Physical change; the substance remains H₂O.
3. Precipitate formation, strong evidence of a chemical reaction.

Reason from Evidence - look for

Single observations can have nonchemical explanations. Multiple consistent signs strengthen the claim that new substances formed.

ACE - Extend look-for

Temperature change and gas production are evidence. Additional evidence might include color change, precipitate, or comparing reactants/products while ensuring the inflation is not just trapped air.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.